

synaptic Technical Reference

synaptic project

Open-source LaTeX community

Abstract: This reference describes synaptic's module graph, load-time invariants, namespace rules, font and colour subsystems, extension points, generated-file policy, automated verification workflow, and packaging details.

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1 Architecture

`synaptic.sty`

```

`-- synaptic-base.sty
    |-- synaptic-lang-{en,zh}.def
    |-- synaptic-color.sty
    |-- synaptic-fonts.sty
    |-- synaptic-layout.sty
    |-- synaptic-theorem.sty
    |-- synaptic-title.sty      shared metadata/renderers
`-- mode module(s)
    |-- synaptic-journal.sty    journal
    |-- synaptic-book.sty       book
    |-- synaptic-lecture.sty    lecture
    |-- synaptic-boxes.sty      lecture
    `-- synaptic-notes.sty      notes

```

The thin shell forwards user options to `synaptic-base` and defines a small set of collision-resistant convenience commands. The base module checks the engine and class, parses options, loads labels, validates book mode, and dispatches core and mode-specific modules. Language definition files are loaded before any module so every label constant is available everywhere.

1.1 Module responsibilities

Module	Kind	Responsibility
<code>synaptic.sty</code>	shell	option forwarding, doc helpers, acknowledgments
<code>synaptic-base.sty</code>	core	engine/class checks, keys, dispatch, setup
<code>synaptic-lang-*.def</code>	data	bilingual label constants
<code>synaptic-color.sty</code>	core	theme registry, semantic tokens, delivery
<code>synaptic-fonts.sty</code>	core	font bundles, auto selection, CJK fonts
<code>synaptic-layout.sty</code>	core	geometry, typography, headings, numbering
<code>synaptic-theorem.sty</code>	core	theorem styles, environments, proofs
<code>synaptic-title.sty</code>	core	metadata model and four title renderers
<code>synaptic-journal.sty</code>	mode	journal navigation and statements
<code>synaptic-book.sty</code>	mode	chapter/part style, matter helpers
<code>synaptic-lecture.sty</code>	mode	masthead, teaching cards, navigation
<code>synaptic-boxes.sty</code>	mode	example/remark/warning/exercise cards
<code>synaptic-notes.sty</code>	mode	notes masthead and knowledge cards

Mode	Class invariant	Extra modules
journal	KOMA article/report	journal
book	chapter-based KOMA class with front matter	book
lecture	KOMA article/report	lecture, boxes
notes	KOMA article/report	notes

2 Configuration lifecycle

The package-level `synaptic` key family is evaluated before module loading. Mode, language, font set, title layout, measure, density, numbering, delivery colour mode, extra math alphabets, and fine-breaking behaviour therefore become structural invariants. The runtime key family exposes only the state that can be applied without reloading packages: `theme`, `toc`, and `toc-layout`. Frozen or unknown keys produce diagnostic warnings.

Book mode is rejected unless both chapter and front-matter interfaces exist. LuaLaTeX and KOMA-Script requirements are fatal diagnostics rather than delayed undefined-control-sequence failures.

3 Design invariants

The following properties are deliberately asserted by the implementation and protected by the regression suite.

- **Deterministic typography.** Font auto-selection and theme derivation never depend on the machine beyond installed fonts; line spacing, measure, and heading sizes are explicit.
- **No global bad-box suppression.** `Emergencystretch` and tolerance are tuned; overfull and underfull warnings remain visible for the author.
- **No forced paper size or page-number reset.** Geometry follows the class; journal titles do not silently restart numbering.
- **One Unicode conversion.** PDF metadata is passed to `hyperref` as source tokens; pre-conversion would double-encode.
- **Collision-resistant public names.** Generic aliases are installed only when free.
- **Semantic colour architecture.** Components consume semantic tokens; themes and delivery modes change tokens, never component logic.

4 Subsystems

4.1 Colour tokens

Built-in themes are stored in a global property list. Activating a theme maps its primary, secondary, and accent colours to semantic public names and derives surface, border, rule, link, readable-muted, decorative-muted, and text tokens. Print and mono delivery modes alter derived roles without changing component code. Danger, warning, and success colours remain semantic constants so their meaning does not change with brand colour.

```
\SynapticDefineTheme{name}{primary}{secondary}[accent]
\SynapticUseTheme{name}
```

The arguments are xcolor expressions. Defining a custom theme registers and activates it.

4.2 Font selection

Font discovery is implemented entirely with `\IfFontExistsTF`. Each candidate is a complete roman/sans/mono/math bundle; auto selection never picks a family whose companions are missing. The fixed priority is XCharter, Libertinus, STIX Two, then Latin Modern. An explicit incomplete choice emits a package error with the missing bundle requirements.

For `lang=zh`, ctex is loaded without a preset font scheme. Noto CJK SC is preferred and Fandol is the TeX Live fallback. Synthetic weight/slant is limited to faces for which the preferred family has no native counterpart.

4.3 Layout and document structure

Geometry combines mode defaults with explicit measure and density profiles. Journal and lecture favour moderate line lengths; notes supports compact single/two-column composition; book uses mirrored inner/outer margins, binding correction, outer running heads, and empty duplex blanks. Widow and club penalties, vertical rhythm, list spacing, line spacing, and paragraph indentation are explicit rather than hidden behind global bad-box suppression.

KOMA heading fonts and skips are installed in the `begindocument/before` hook, with all numbered levels sharing the sans heading family: the primary colour for `section`, dark for `subsection`, and a smaller, muted tint for `subsubsection` so the depth hierarchy stays legible. Hyperref, bookmark, microtype, captions, lists, caption punctuation, and mode-aware equation numbering are configured centrally. A hairline running-head separator (`headsepline`) is installed by the mode modules through a shared helper and coloured by the dedicated KOMA `headsepline` font, keeping decoration lighter than head text.

The table of contents is refined through KOMA's `\DeclareTOCStyleEntry` immediately before it is typeset: top-level entries repeat the on-page heading colour in bold, deeper entries remain dark, and number alignment and vertical rhythm are tightened. Figure and table caption labels carry the secondary theme token.

4.4 Titles and metadata

Title values are held in `expl3` token lists and sequences. Authors and their affiliation marks use parallel sequences; affiliations and emails are repeatable. A common metadata model feeds four renderers: inline journal, book title sequence, course masthead, and compact notes masthead. The title command validates a non-empty title and controls optional TOC placement without silently resetting the user's page numbering. Standard LaTeX metadata is an input adapter, not a redefinition of `\maketitle`.

PDF information is passed as source tokens to `hyperref`. `Hyperref` performs the Unicode PDF-string conversion once; pre-converting and then calling `\hypersetup` would double-encode metadata.

4.5 Theorems and boxes

`thmtools` defines plain, definition, and remark styles on top of `amsthm`. The built-in theorem family (`theorem`, `lemma`, `proposition`, `definition`, `corollary`, `axiom`, `conjecture`, `claim`, `property`, `criterion`, `convention`, `problem`, `solution`, `example`, `remark`, and `note`) uses a mode-aware shared counter: chapter scope for books and section scope elsewhere by default. Under `theorem-style=boxed` the numbered statement environments are wrapped with `tcolorbox` (a theme-tinted background with a west accent bar) at `\begindocument`; proofs stay unboxed. `theorem-style=plain` disables that wrapping. Lecture mode adds its box-card environments; examples and exercises can share the theorem counter, while warning and remark cards remain unnumbered by default. Public short names use the `syn` prefix; v2.0 generic aliases are conditional compatibility shims. In the separate numbering profile, `tcolorbox` theorem counters follow the `tcb@cnt@<environment>` naming convention.

5 Namespace policy

Scope	Pattern	Example
Public command	<code>\Synaptic...</code>	<code>\SynapticDefineTheme</code>
Public environment	<code>syn...</code>	<code>synwarning</code>
Internal function	<code>__synaptic_...:</code>	title rendering helper
Module interface	<code>\synaptic_...:</code>	mode accessor
Local state	<code>\l_synaptic_...</code>	selected theme token list
Global registry	<code>\g_synaptic_...</code>	theme property list
Label constant	<code>\c_synaptic_label_...</code>	theorem label

New public names should avoid common nouns. Compatibility aliases must use `\ProvideDocumentCommand` or an explicit existence check.

6 Extension points

- Add labels in a new `synaptic-lang-*.def` file, then extend the language key and definition-file dispatch.
- Add a complete font predicate, setter, explicit dispatch branch, and a deliberate location in the auto-selection chain.
- Add modes in the base key family, layout dispatch, and mode-module loader. State the required KOMA class capability and test it.
- Build visual extensions from semantic colour tokens rather than fixed RGB values.
- Add tests along with every extension; the suite is the guard for the invariants above.

7 Automated verification

`synaptic.dtx` is the source of truth. DocStrip uses `synaptic.ins` to generate the installable `.sty` and `.def` files. The generated copies under `tex/latex/synaptic/` are committed for zero-install use and must remain byte-identical to fresh extraction.

The eighteen-group regression suite is organised as follows:

Group	Coverage
journal, lecture, notes, book	core mode behaviour, navigation, and defaults
book-zh, lecture-zh, journal-zh, language-zh	Chinese labels, CJK fonts, and bilingual layering
themes	cycling all five themes and a custom theme
color-mode-print, options	print and monochrome delivery
customization	runtime setup, custom theorems
toc-layouts, title-layouts	TOC and title compositions
layout-profiles	density, measure, binding offset
book-full	complete title sequence and matter workflow
lecture-continuous	continuous and separate numbering profiles
extra-math	optional Unicode math alphabets

Each group typesets a document with the standard engine and compares the normalised log against its committed expectation file. CI runs the suite and additionally builds the documented source, four manuals, and all examples, failing on warnings, overfull boxes, or missing characters.

8 Source and release workflow

```

13build unpack      # regenerate .sty/.def into build/local
13build check       # run the regression suite
13build doc         # build the documented source, manuals
13build ctan        # create TDS and CTAN archives
13build install     # copy package files into the texmf tree
13build uninstall  # remove a previous installation

```

The `ctan` target emits `synaptic-ctan.zip` (the source tree plus generated documentation and examples) together with the nested `synaptic.tds.zip` in TDS layout. Installation replaces the previous version in place; because the generated file set is stable, `13build install` is safe to rerun after every release.

9 Compatibility notes

Version 2.3 retains the v2.2 API unchanged. Lower-case book matter helpers and generic lecture box names remain conditional v2.0 aliases, the `theme` key remains the palette selection interface, and new documents should use the namespaced public forms. Removed `style=` and `color=` options remain explicit errors with migration hints.